11:53 4G 6 K/s 86

quiz.vu.edu.pk/QuizQue: 3

Question # 10 of 10 (Start time: 11:53:14 AM, 03 June 2021)

If the sequence is convergent then

Select the correct option

- it is bounded below but may not be bounded above.
- it is bounded above but may not be bounded below.
- it has two limits.
- it is bounded.

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11:52 4G 13 K/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 89 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 9 of 10 (Start time: 11:52:19 AM, 03 June 2021)

Total Marks: 1

The series $\sum_{n=0}^{\infty} r^n, \quad -1 < r < 1$, is

Select the correct option

Reload Math Equations

- divergent.
- convergent and converges to $\frac{1}{1-r}$.
- convergent and converges to 5.
- convergent and converges to $\frac{1}{1+r^2}$.

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Click to Save Answer & Move to Next Question

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11:51 9 K/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 89 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 8 of 10 (Start time: 11:51:23 AM, 03 June 2021)

Total Marks: 1

If $f(x) = \sqrt{4 - x^2}$ and $g(x) = \sqrt{x - 1}$, then domain of the quotient function f/g is

Select the correct option

Reload Math Equations

- $(1, 2]$.
- $[1, 2]$.
- $[7, 8]$.
- $[-3, 3]$.

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Click to Save Answer & Move to Next Question

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11:49 4G 14 K/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 89 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 7 of 10 (Start time: 11:49:33 AM, 03 June 2021)

Total Marks: 1

$If\ \{s_n\}\ is\ ------,\ then\ \lim_{n\to\infty} s_n = \inf\{s_n\}.$

Select the correct option

Reload Math Equations

- nonincreasing
- increasing
- decreasing
- nondecreasing

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Click to Save Answer & Move to Next Question

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11:48 4G 18 K/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 89 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 6 of 10 (Start time: 11:48:19 AM, 03 June 2021)

Total Marks: 1

The value of the $\lim_{n\to\infty} \frac{1}{n^p}$ is

Select the correct option

Reload Math Equations

- $\lim_{n\to\infty} \frac{1}{n^p} = 3$ for all p.
- $\lim_{n\to\infty} \frac{1}{n^p} = 0$ for $p > 0$.
- $\lim_{n\to\infty} \frac{1}{n^p} = 1$ for all $p > 1$.
- $\lim_{n\to\infty} \frac{1}{n^p} = 1$ for all p.

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Click to Save Answer & Move to Next Question

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11:47 258 B/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

MTH621:Quiz-1 Quiz Start Time: 1

Question # 5 of 10 (Start time: 11:47:23 AM, 03 June 2021)

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\bar{s}$ such that ,if $\varepsilon > 0$, $s_n < \bar{s} + \varepsilon$ - - - - -

Select the correct option

- *for small n*
- *for large n*

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11:46

quiz.vu.edu.pk/QuizQue:

3

MTH621:Quiz-1

Question # 4 of 10 (Start time: 11:46:14 AM, 03 June 2021)

The series $$\sum_{n=0}^\infty r^n,\quad-1

Select the correct option

- divergent.
- convergent and converges to $\frac{1}{1+r^2}$.
- convergent and converges to 5.
- convergent and converges to $\frac{1}{1-r}$.

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11:44 4G 7 K/s 87

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 88 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 3 of 10 (Start time: 11:44:53 AM, 03 June 2021)

Total Marks: 1

The value of the

$$\lim_{n\to\infty} \frac{1}{n^p}$$

is

Select the correct option

Reload Math Equations

- $\lim_{n\to\infty} \frac{1}{n^p} = 0$ for $p > 1$.
- $\lim_{n\to\infty} \frac{1}{n^p} = 3$ for all p.
- $\lim_{n\to\infty} \frac{1}{n^p} = 1$ for all p.
- $\lim_{n\to\infty} \frac{1}{n^p} = 1$ for $p > 1$.

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Click to Save Answer & Move to Next Question

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quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 88 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2021

Question # 2 of 10 (Start time: 11:44:14 AM, 03 June 2021)

Total Marks: 1

Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite.Then $\sum_{n=k}^{\infty} ca_n =$ - - - - - -

Select the correct option

Reload Math Equations

- $c - A$
- $c + A$
- cA
- $\frac{c}{A}$

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Click to Save Answer & Move to Next Question

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11:42 44 K/s 88

quiz.vu.edu.pk/QuizQue: 3

MC200402273: TAJAMMAL HUSSAIN

Time Left 88 sec(s)

MTH621:Quiz-1

Quiz Start Time: 11:42 AM, 03 June 2

Question # 1 of 10 (Start time: 11:42:35 AM, 03 June 2021)

Total Mark

If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol $\sum_{n=k}^{\infty} a_n$ is - - - - - - - - - and a_n is the nth term of series.

Select the correct option

Reload Math Equatio

infinite series

S KHAN ACADEMY

finite series

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 09:14:01 AM, 03 June 2021)

Which statement(s) is(are) true for the following sequence $\lim_{n\to\infty} \sqrt[n]{n}$

Select the correct option

- The sequence is divergent to
- The sequence converges to 0
- The sequence converges to 1
- The sequence is oscillatory

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Question # 8 of 10 (Start time: 09:11:57 AM, 03 June 2021)

Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite.Then $\sum_{n=k}^{\infty} ca_n =$ - - - - - -

Select the correct option

- $c - A$
- cA
- $c + A$
- $\frac{c}{A}$

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AJLAL ABBAS

Time Left 85 sec(s)

Quiz Start Time: 09:05 AM, 03 June 20

0 (Start time: 09:12:47 AM, 03 June 2021)

Total Marks

$If\ \{x_n\}\ is\ ----,\ then\ \{x_n\}\ has\ a\ convergent\ subsequence.$

t option

Reload Math Equations

unbounded

bounded

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9:12 AM

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Quiz St

Question # 7 of 10 (Start time: 09:11:22 AM, 03 June 2021)

$For \ ------series \ \sum_{n=1}^{\infty} \frac{1}{n} = 0.$

Select the correct option

- None of these
- arithmetic
- harmonic
- geometric

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MTH621:Quiz-1

Question # 4 of 10 (Start time: 09:08:24 AM, 03 June 2021)

If $\sum a_n$ ------, *then*

Select the correct option

- $\lim_{n \to \infty} a_n = 0$
- $\lim_{n \to \infty} a_n = -1$
- None of these
- $\lim_{n \to \infty} a_n = 1$

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BC180408070: AJLAL ABBAS

MTH621:Quiz-1

Quiz Sta

Question # 6 of 10 (Start time: 09:10:13 AM, 03 June 2021)

For the convergent sequence

$\{s_n\}$,

which statement is true?

Select the correct option

- $\limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = \lim_{n\to\infty} s_n$.
- has a divergent subsequence.
- $\limsup_{n\to\infty} s_n \neq \liminf_{n\to\infty} s_n = \lim_{n\to\infty} s_n$.
- $\limsup_{n\to\infty} s_n \neq \liminf_{n\to\infty} s_n \neq \lim_{n\to\infty} s_n$.

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MTH621:Quiz-1

Quiz Start Time: 09:

Question # 5 of 10 (Start time: 09:09:09 AM, 03 June 2021)

$If\ |s_n| \leqslant r\ for\ any\ sequence\ \{s_n\},\ where\ r\ is\ a\ real\ number,\ then\ \{s_n\}\ is\ -------$

Select the correct option

Reload

- None of these
- S KHAN ACADEMY
- *Bounded sequence*
- *Bounded below*
- *Bounded above*

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Quiz Start Ti

Question # 3 of 10 (Start time: 09:07:26 AM, 03 June 2021)

If $s_n \leqslant b$ for anysequence $\{s_n\}$, where b is a real number, then $\{s_n\}$ is - - - - - - -

Select the correct option

- Bounded above
- Bounded sequence
- None of these
- Bounded below

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BC180408070: AJLAL ABBAS

MTH621:Quiz-1

Question # 2 of 10 (Start time: 09:06:18 AM, 03 June 2021)

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\overline{s}$ such that
, if $\varepsilon > 0$, $s_n < \overline{s} + \varepsilon$ ------

Select the correct option

- *for small n*
- *for large n*

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BC180408070: AJLAL ABBAS

MTH621:Quiz-1

Question # 1 of 10 (Start time: 09:05:24 AM, 03 June 2021)

A point $\bar{x}$ is a limit point of a set iff there is a sequence $\{x_n\}$ of point in S such that $x_n \neq x$ for $n \geq 1$, and - - - - - - -

Select the correct option

- $\lim_{n \to \infty} x_n = \bar{x}$
- $\lim_{n \to \infty} x_n \neq \bar{x}$
- None of these
- $\lim_{n \to \infty} x_n > \bar{x}$

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June 2021)

The sum of a convergent series is ------.

different

unique

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Question # 9 of 10 (Start time: 11:06:24 AM, 03 June 2021)

The given series $\sum_{n=1}^{\infty} \frac{1}{n}$ is ------.

Select the correct option

- divergent
- convergent

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300: ASIF ALI

uiz-1

8 of 10 (Start time: 11:05:18 AM, 03 June 2021)

If $|x| < 1$, then

the correct option

$\lim_{n\to\infty} x^n = 1$

$\lim_{n\to\infty} x^n = 0$

$\lim_{n\to\infty} x^n = -1$

None of these

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n # 7 of 10 (Start time: 11:04:35 AM, 03 June 2021)

A divergent infinite series that does not diverge to $\pm\infty$ is said to be ------.

t the correct option

Reload M

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oscillate

convergent

None of these

bounded

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on # 6 of 10 (Start time: 11:03:21 AM, 03 June 2021)

sec(s)

Quiz Start Time: 10:58 AM, 03 June 2021

Total Marks: 1

If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol $a_1 + a_2 + a_3 + \cdots + a_n + \cdots$ is an infinite series and a_n is the nth term of series.

ct the correct option

- $\prod_k^\infty a_n$
- None of these
- a_n
- $\sum_{n=k}^{\infty} a_n$

Reload Math Equations

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MTH621:Quiz-1

Question # 5 of 10 (Start time: 11:02:30 AM, 03 June 2021)

Quiz Start Time: 10:

Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite.

Then $\sum_{n=k}^{\infty} (a_n + b_n) =$ ------

Select the correct option

- $A - B$
- AB
- $A + B$
- $\frac{A}{B}$

Reload

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1621 Quiz-1

Quiz Start Time: 10:58 AM, 03

estion # 4 of 10 (Start time: 11:01:37 AM, 03 June 2021)

Total

If $\{x_n\}$ unbounded below, then $\{x_n\}$ has a subsequence $\{x_n\}$ such that $\lim_{k\to\infty} x_{n_k} = - - - -$

Select the correct option

Reload Math Equat

- $-\infty$
- 0
- ∞
- 1

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estion # 2 of 10 (Start time: 10:59:59 AM, 03 June 2021)

The limit of the sequence

$$s_n = \frac{1}{n} + \frac{2(1+3/n)}{1+1/n}.$$

Select the correct option

Reloa

- 2
- 4
- 1
- 3

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Every Cauchy sequence has a

Select the correct option

- decreasing subsequence.
- increasing subsequence.
- convergent subsequence.
- positive subsequence.

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